ADVANCED MACHINE LEARNING · WORKSHEET 01

The ML Project Lifecycle (Part 1)

Problem Definition · Data Collection · Exploratory Analysis · Preprocessing

Lecture 1

Name:	**Branch / Year:**	**Date:**

THE ML PROJECT LIFECYCLE

PART 1 DEFINING THE PROBLEM (THE FOUNDATION)

▷ **HOOK**

Your music streaming company says: "Users are leaving! Build an AI to fix this!" If you open a laptop and start coding a linear regression right now, why are you almost guaranteed to fail?

The five-step blueprint

An algorithm cannot guess what your business needs. Phase 1 translates a messy human problem into an **explicit mathematical framework** using five steps:

Step	**What You Do**	**Example**
1. Understand the Problem	Isolate the exact friction point	Users canceling premium subscriptions = churn
2. Set Quantifiable Goals	Define a measurable target	"Reduce churn by 10% over two quarters"
3. Assess ML Feasibility	Ask: do we need ML? Do we have data?	Can a SQL query solve it? If yes, skip ML.
4. Identify Constraints	Budget, latency, privacy laws (GDPR)	Predictions in <10 ms on a phone?
5. Stakeholder Alignment	PMs, devs, execs agree on success	Shared definition of "success"

Blueprint in action

- **Vague:** "We want to improve our online retail store."
- **Rigorous:** "Using historical clickstreams and purchase data, predict which users will stop shopping within 3 months."

★ KEY INSIGHT

A poorly defined problem wastes more time than a bad algorithm. You cannot optimize “make it better” the algorithm needs a number to minimize.

▷ PRACTICE P1 · Rewrite each vague request as a rigorous ML problem statement

(a) “Our hospital needs better patient care.”
→ ____________________

(b) “Make our email system smarter.”
→ ____________________

△ REFLECT 1

A company has excellent customer data, but it is not permitted to use the data for training an ML model. At which phase of the Machine Learning Blueprint does the project encounter this problem?

Hint: Operational Constraints.

⧫ TAKEAWAY

Never touch code until you have a quantified target, confirmed data access, and stakeholder agreement.

PART 2 DATA COLLECTION (SOURCING THE FUEL)

▷ HOOK

Can you build an ML model without any data? An algorithm has no concept of reality it only knows the world you present via training data. Flawed collection = broken model.

Five production data sources

Source	Example
Internal Warehouses (SQL databases)	Customer purchase history, transactional logs
Public Repositories	Kaggle, UCI ML Repository
APIs	Live stock prices, weather feeds
Web Scraping	Product prices from competitor sites
Manual Labeling	Radiologists labeling tumors on X-rays

Data structures

Type	Structure	Example
Structured	Rigid rows & columns (SQL)	Order ID, Timestamp, Amount, User ID
Unstructured	No grid format	Images, audio, video, free-text reviews
Semi-structured	Tags/keys but no strict schema	JSON, XML, IoT sensor logs
Time-series	Indexed over time intervals	Daily stock prices, hourly weather data

Feature types

Type	Nature	Example
Categorical (Nominal)	Labels, no order	Color, City, Blood Type (A/B/AB/O)
Categorical (Ordinal)	Labels with rank	Education: HS < Bachelor < Master < PhD
Numerical (Discrete)	Countable integers	Number of rooms, items in cart
Numerical (Continuous)	Infinitely divisible	Income, temperature, weight

★ KEY INSIGHT

An algorithm only knows the world you show it. Garbage In = Garbage Out.

▷ PRACTICE P2 · Classify each feature: Structured / Unstructured / Semi / Time-Series and Categorical / Numerical

(a) A food delivery app stores customer ratings as 1, 2, 3, 4, or 5 stars.
Structure: Structured Feature: Categorical (Ordinal)

(b) A hospital records patient body temperature every hour for 7 days after surgery.
Structure: Structured + Time series Feature: Numerical (Continuous)

(c) An e-commerce platform stores each product's description written by the seller in free text.
Structure: Unstructured Feature: Textual

(d) A government database stores the PIN code (postal code) of each citizen's address.
Structure: Str. Feature: Categ (Nominal)

(e) A weather station logs wind speed in km/h every 10 minutes for an entire year.
Structure: St + Time Series Feature: Numer (Continuous)

△ REFLECT 2

Your scraper collects 100k reviews but 40% are bot-generated spam. Is more data always better?
Hint: quality vs. quantity.

♦ TAKEAWAY

Know your sources, know the structure, know the feature types before you write a single line of code.

PART 3 EXPLORATORY DATA ANALYSIS (EDA)

▷ HOOK

Your company hands you 500,000 customer records. What is the very first thing you do? If you said "train a model" stop. A doctor does not prescribe treatment before examining the patient.

EDA is the **diagnostic stage**. It transforms raw data into actionable insights before any preprocessing.

EDA checklist

Goal	What You Check
1. Dataset Structure	Rows, columns, data types, feature names
2. Missing Values	Which columns? How many? Random or systematic?
3. Outliers	Extreme values (data error? fraud? rare case?)
4. Duplicates	Repeated rows from collection/merging errors
5. Distributions	Normal (bell) vs. Skewed (left/right)
6. Relationships	Correlation, dependency, redundant features
7. Class Imbalance	95% Non-Fraud vs. 5% Fraud → accuracy lies

Analysis levels

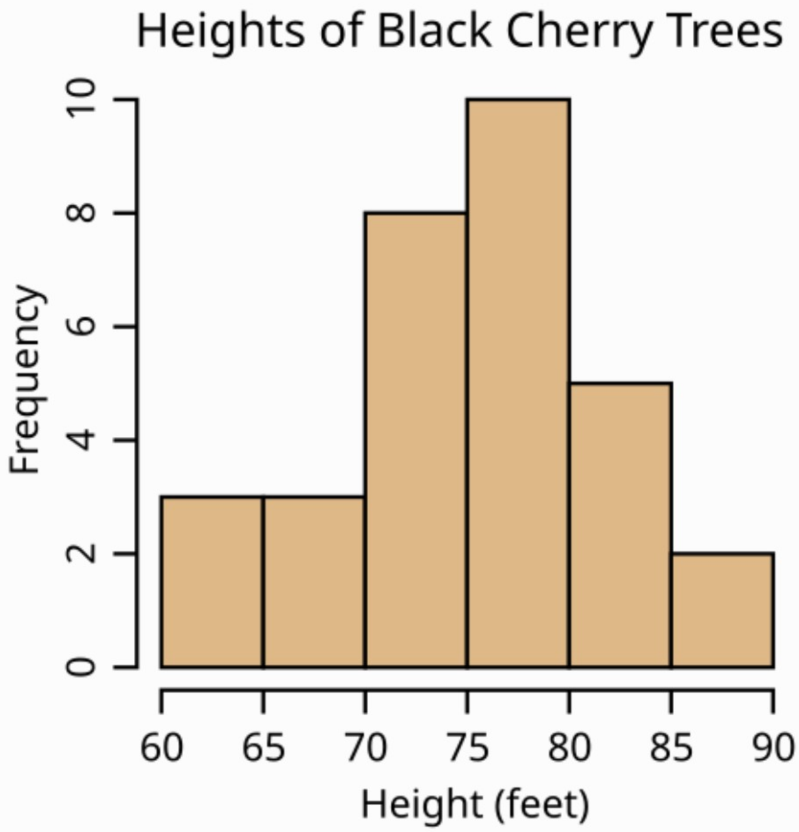
- **Univariate:** One variable (distribution, mean, median, IQR, outliers). Tool: Histogram, Box Plot.
- **Bivariate:** Two variables (correlation, trends). Tool: Scatter Plot, Correlation Heatmap.
- **Multivariate:** Many variables (interaction effects). Tool: Heatmap, Pair Plot.

Common visualizations

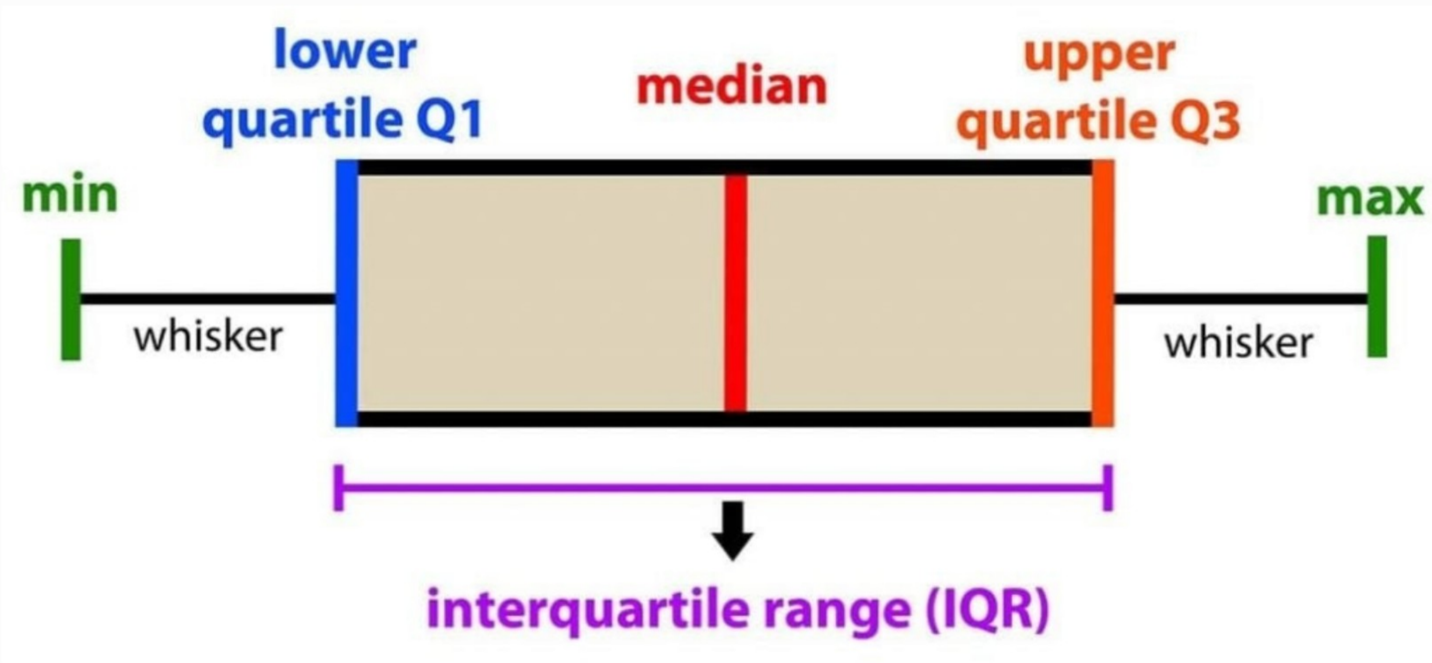
Chart	Use For
Histogram	Distributions, skewness
Box Plot	Outliers, spread (Q_1, Q_3, IQR)
Bar Chart	Categorical comparisons
Scatter Plot	Relationship between two numerical variables
Correlation Heatmap	Feature correlations ($+1$, 0, -1)

Correlation

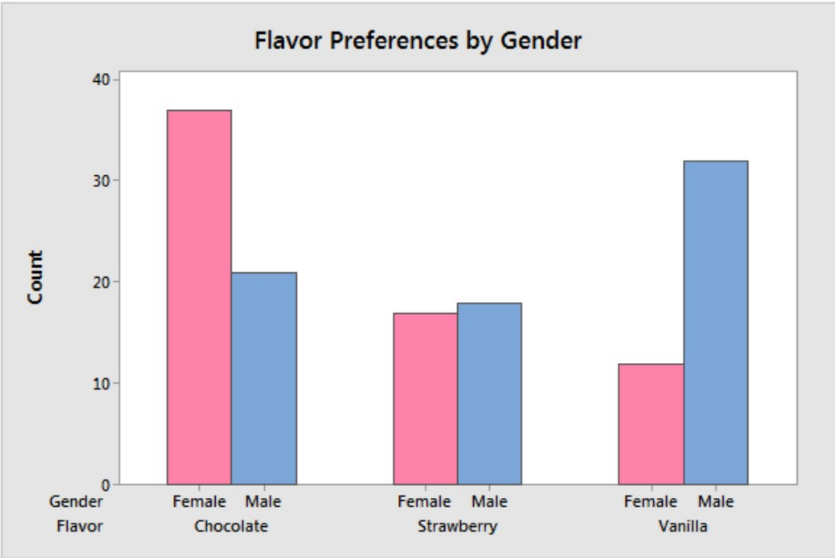
- **Positive:** both increase together (Experience ↔ Salary)
- **Negative:** one up, other down (Price ↔ Demand)
- **None:** no clear relationship



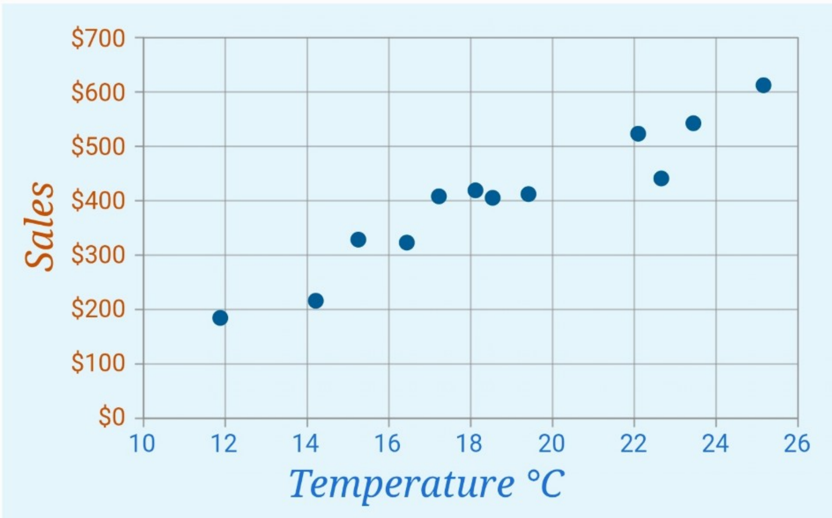
Histogram



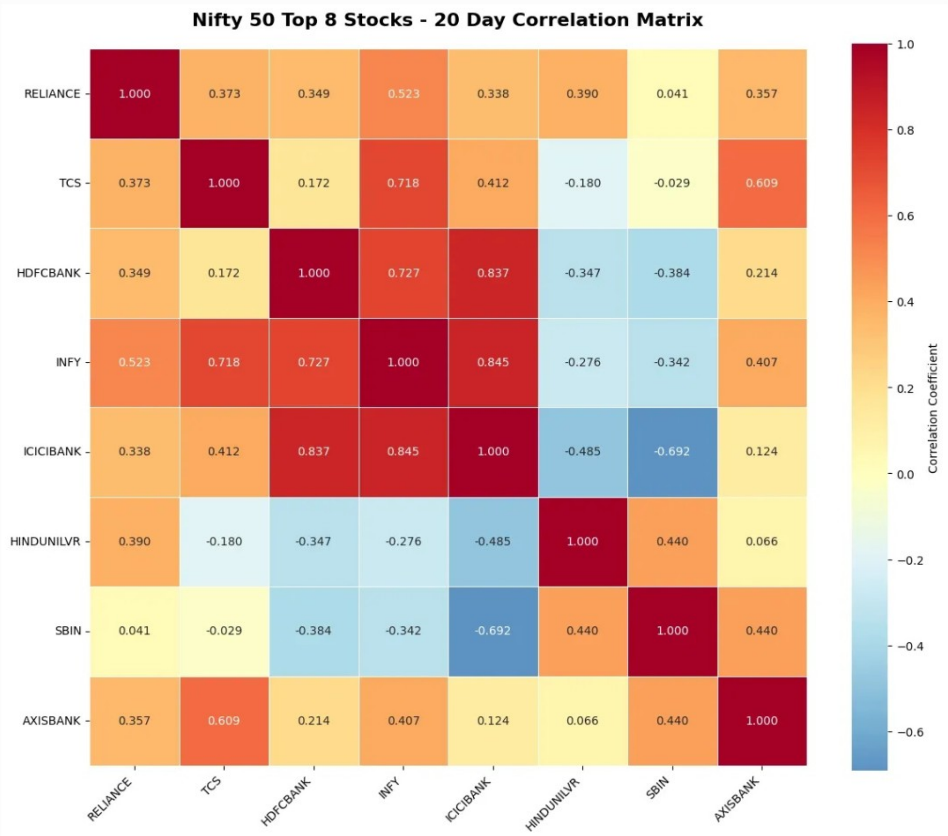
↳ Box Plot



Bar Chart



→ Scatter Plot



→ Heatmap

★ KEY INSIGHT

Skip EDA and you risk training on missing values, outliers, duplicates, and imbalanced classes all of which silently destroy performance. EDA is not optional.

▷ PRACTICE P3 · Given this dataset, list your EDA steps and pick the right chart

Dataset: 100,000 rows columns: Customer_ID, Age, Gender, Annual_Income, Spending_Score, City.

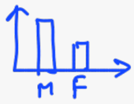
First 3 EDA steps you perform:

1. ____
2. ____
3. ____

Chart for Age distribution? Histogram

Chart for Income vs. Spending_Score? Scatter Plot

Chart to check if Gender affects Spending_Score? Bar Chart



△ REFLECT 3

A dataset is 95% Class A and 5% Class B. A model predicts A every time, scoring 95% accuracy. Is this useful?

Hint: what is the recall for Class B?

◆ TAKEAWAY

EDA first, algorithm later. Understand structure → find issues → visualize patterns → then preprocess.

PART 4 DATA PREPROCESSING (THE REAL ENGINEERING WORK)

▷ HOOK

In textbooks, data is always clean. In the real world? Messy, biased, incomplete. Can a sophisticated algorithm self-correct and ignore the garbage? No. Garbage In, Garbage Out.

Algorithms see data as matrices of numbers. Missing value? Equation breaks. One feature in millions, another in single digits? The large one overwhelms the math. Preprocessing builds a **clean numerical matrix**.

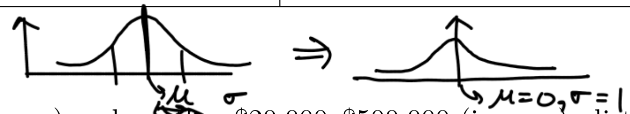
Step 1: Handling missing values

Strategy	Action	When to Use	Danger
Deletion	Drop rows or columns	<1–2% missing; or column >80% empty	Discards info, introduces bias
Imputation	Fill with mean/median (num) or mode (cat)	Moderate missingness	May introduce artificial patterns

Step 2: Handling inconsistent data

Type	Issue	Fix
Mixed Units	Weight in both kg and lbs	Standardize to single unit
Typos	'Californa', 'California', 'CA'	Regex / Levenshtein distance matching
Duplicates	Same record logged twice	Remove duplicate records
Inconsistent Labels	Yes, Y, True, 1 all mean "yes"	Map all to single label format

Step 3: Feature scaling



When one feature ranges from 1–5 (bedrooms) and another \$20,000–\$500,000 (income), distance-based algorithms (KNN, SVM) and gradient descent are dominated by the larger feature. Scaling fixes this.

Technique	Formula	Use When	Risk
A. Min-Max (Normalization)	$X_{\text{scaled}} = \dfrac{X - X_{\min}}{X_{\max} - X_{\min}}$ Maps to [0, 1]	Not Gaussian; neural nets; bounded ranges	One outlier squashes everything near 0.
B. Z-Score (Standardization)	$Z = \dfrac{X - \mu}{\sigma}$ Mean→0, Std→1	Data follows Normal distribution	Performs best when features are approximately Gaussian.
$\boxed{\text{C. Max Absolute}}$	$X_{\text{scaled}} = \dfrac{X}{\max(\lvert X\rvert)}$ Maps to [−1, 1]	Sparse data (many zeros, e.g., bag-of-words)	Preserves zeros and sign
D. Robust	$X_{\text{scaled}} = \dfrac{X - \text{Median}}{Q_3 - Q_1}$ $\text{IQR} = Q_3 - Q_1$	Heavy outliers you cannot drop	Ignores mean/std; uses median + middle 50%

★ KEY INSIGHT

Min-Max is sensitive to outliers (one extreme value squashes everything). Z-Score assumes normality. Robust Scaling is your safety net when outliers are unavoidable.

▷ PRACTICE P4 · Compute by hand

A. MIN-MAX SCALING

Dataset $X = [10, 20, 30, 40, 50]$. → $x_{max} = 50$, $x_{min} = 10$

Scale $X = 10$: $(10-10)/(50-10)$ Scale $X = 30$: 0.5 Scale $X = 50$: ______

B. Z-SCORE STANDARDIZATION

Dataset $X = [2, 4, 6, 8, 10]$ → $\mu = 6$, $\sigma = \sqrt{8} = \sqrt{\frac{1}{n}\sum (x_i - \bar{x})^2}$

z for $X = 2$: $(2-6)/\sqrt{8}$ z for $X = 6$: ______ z for $X = 10$: ______

C. ROBUST SCALING (full walkthrough) *(Optional)*

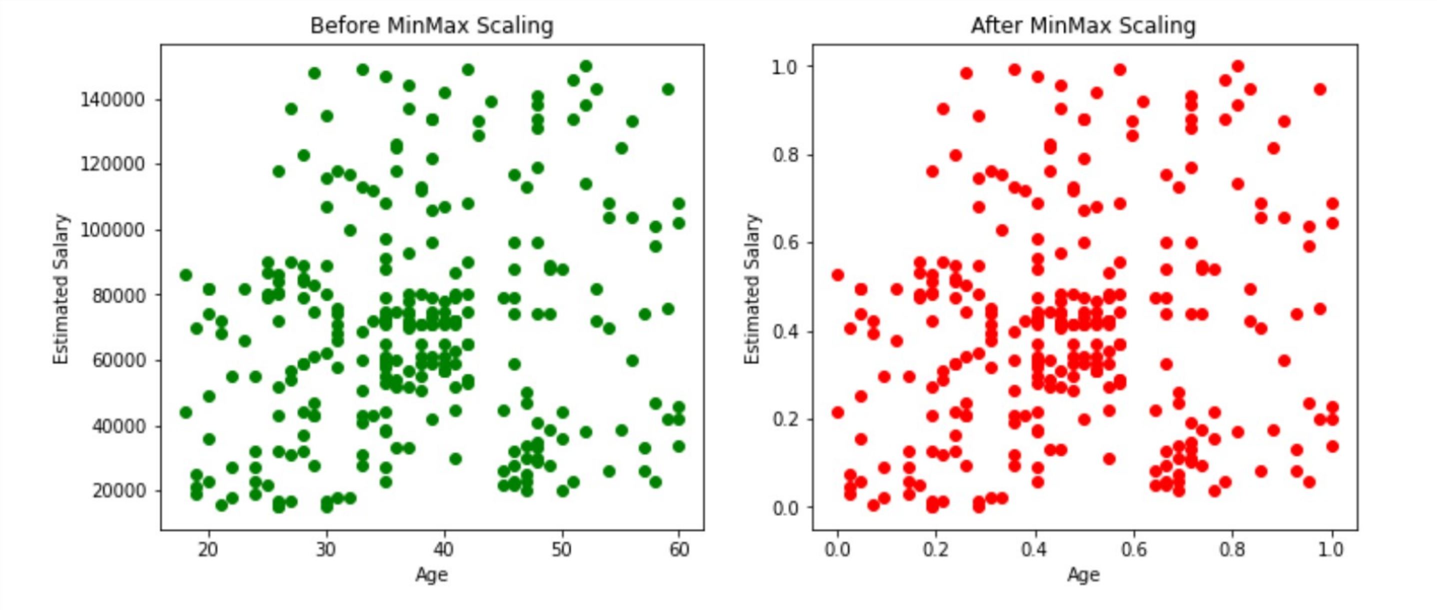
Dataset $X = [100, 150, 200, 250, 50000]$.

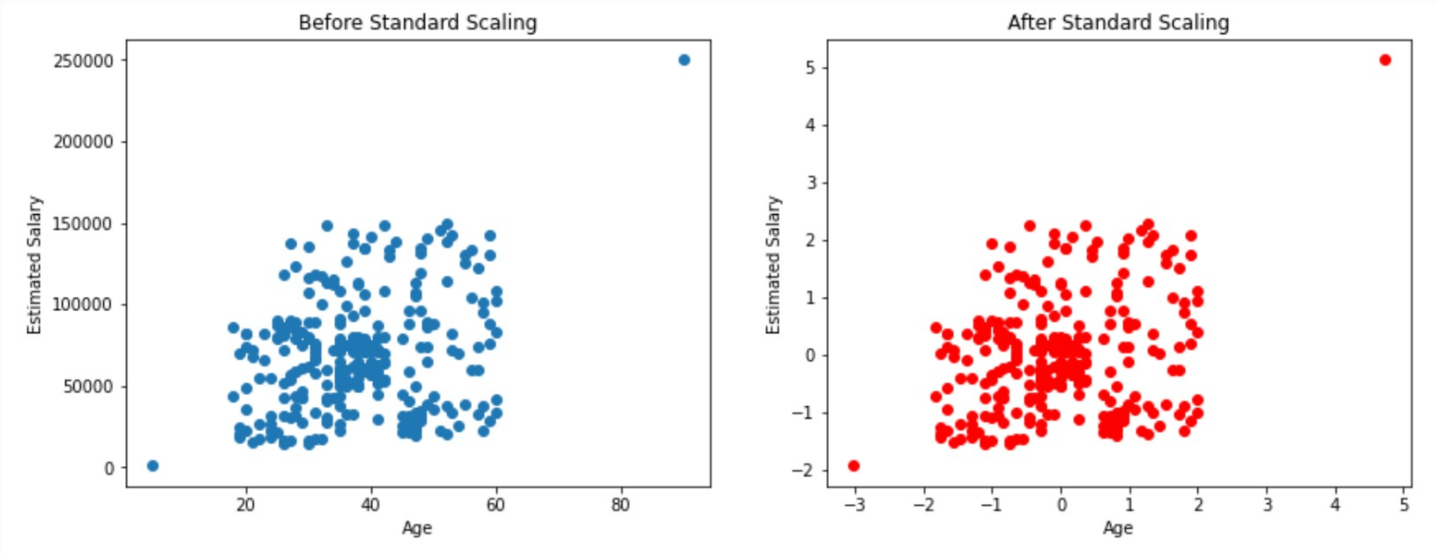
Step 1 Sort: [______, ______, ______, ______, ______]
Step 2 Median (Q_2): 200
Step 3 Q_1 (median of lower half [100, 150]): 125
Step 4 Q_3 (median of upper half [250, 50000]): $(250 + 50000)/2 = 25125$

Customer ID	F/NF
	NF
	F
	NF
	NF
	F
	NF
	⋮

SMOTE

F → 2 NF → 1000


Before MinMax Scaling
After MinMax Scaling
Estimated Salary
Age
140000
120000
100000
80000
60000
40000
20000
20
30
40
50
60
1.0
0.8
0.6
0.4
0.2
0.0


Before Standard Scaling
After Standard Scaling
Estimated Salary
Age
250000
200000
150000
100000
50000
0
20
40
60
80
5
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3
2
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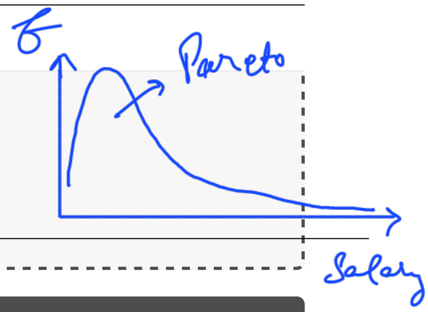
0
−1
−2
−3



Step 5 IQR $= Q_3 - Q_1 =$ ________ $-$ ________ $=$ 25000

Step 6 Scale $X = 100$: $(100-$ 200 $)$ / 25000 $=$ -1/250

Step 7 Scale $X = 50000$: $(50000-$ ________ $)$ / ________ $=$ ________

Why didn't the outlier (50000) break the scale? ____________________

▷ PRACTICE P5 · Pick the scaler and justify in one line

(a) Salary data: \$20k–\$5M with CEO outliers at \$50M.
Scaler: Robust Why: ____________________

(b) Pixel values for images, range $[0, 255]$, no outliers.
Scaler: Min-Max Scaling Why: ____________________

△ REFLECT 4

Why does Min-Max scaling fail when one value is 50,000 and the rest are under 300?
Hint: what happens to $(X_{\max} - X_{\min})$ *in the denominator?*

◆ TAKEAWAY

Match the scaler to the data: Min-Max for bounded/clean, Z-Score for Gaussian, Robust for outliers, Max Abs for sparse.

Step 4: Categorical encoding

If we map Red=0, Green=1, Blue=2, the model thinks Blue > Green > Red, and that (Red+Blue)/2 = Green. That is a **mathematical lie**. Choose encoding carefully.

Encoding	How	Safe For	Danger
Label	Unique integer per category Red=0, Blue=1, Green=2	Binary targets (Spam=1/0); tree-based models (split, not distance)	Implies false order for distance-based models (KNN, SVM)
Ordinal	Controlled rank mapping Low=0, Med=1, High=2	Categories with natural rank; tree + linear models	Wrong if categories are actually unordered
One-Hot	N binary columns (1=present) Red→[0,0,1], Blue→[1,0,0]	Unordered nominal data with few unique values	Curse of Dimensionality with many categories
Binary	Integer → binary digits → split across $\log_2 N$ columns	Many categories; space-efficient alternative	Lower dimensional representation than One-Hot

One-Hot worked: R

Input	Color_Blue	Color_Green	Color_Red
Red	0	0	1
Blue	1	0	0
Green	0	1	0
Red	0	0	1

1
2
3
4
⋮

↳ 1024 categories
↳ One Hot → 1024 → 2^{10} columns

Binary worked (Blue=1→01, Red=2→10, Green=3→11):

$\log_2 1024 = 10$

Input	Bit_0	Bit_1
Red (10)	1	0
Green (11)	1	1
Blue (01)	0	1

0 0 0 0 0 0 0 0 0 1
1 0 0 0 0 0 0 0 0 0
↳ 2^{10}

★ KEY INSIGHT

One-Hot: 3 categories = 3 columns. Binary: 3 categories = 2 columns. For 1000 categories, One-Hot = 1000 cols; Binary ≈ 10 cols.

▷ PRACTICE P6 · Pick the encoding and justify

(a) 500 city names, algorithm = Logistic Regression.
Encoding: Binary Encoding Why: ____

(b) Education (HS < Bachelor < Master < PhD), algorithm = Linear Regression.
Encoding: Ordinal Why: ____

(c) Color (Red, Blue, Green), algorithm = KNN.
Encoding: One hot Why: ____

▷ PRACTICE P7 · Binary encoding by hand

Categories: [Cat, Dog, Fish, Bird]. Assign integers 1–4, convert to binary, fill the table.

Cat = 1 → binary 0 0 1 Dog = 2 → binary 0 1 0
Fish = 3 → binary 0 1 1 Bird = 4 → binary 1 0 0

How many bits needed? 3 How many columns would One-Hot need? 4

△ REFLECT 5

Why can Label Encoding mislead KNN but not a Decision Tree?

Hint: think about distances vs. splits don't worry, this will be covered later.

♦ TAKEAWAY

Ordinal for ranked categories, One-Hot for nominal with few values, Binary for nominal with many. Label only for tree models or binary targets.

SUMMARY OF THE LECTURE

#	Phase	Key Idea
1	Define the Problem	Vague goal → rigorous ML objective
2	Collect Data	Internal DBs, APIs, scraping, labeling
3	Exploratory Data Analysis	Structure, missing, outliers, distributions
4	Data Preprocessing	Clean, scale, encode → numerical matrix

BY THE END OF THIS SHEET YOU CAN

- ☐ Translate a vague business request into a constrained ML problem statement with quantifiable goals.
- ☐ Classify data sources and distinguish structured, unstructured, semi-structured, and time-series data.
- ☐ Perform EDA: detect missing values, outliers, duplicates, class imbalance, and interpret correlations.
- ☐ Select and compute the correct feature scaling (Min-Max, Z-Score, Robust) by hand.
- ☐ Choose the right categorical encoding (Label, Ordinal, One-Hot, Binary) and justify your choice.